Are Contagion Hubs Causal? A Calibrated Counterfactual Test of Stablecoin Network Centrality

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Abstract

When a stablecoin breaks its dollar peg and the panic spreads, analysts routinely use network centrality, or, increasingly, graph neural networks, to name the "hubs" that drive the contagion, and regulators may act on that ranking. But a venue that *moves with* a crisis is not necessarily one that *causes* it, and acting on a merely correlational ranking can waste a scarce intervention budget on a venue that transmits nothing. This is a causal question, and centrality alone cannot answer it. We build a calibrated, intervenable model of stablecoin contagion that answers the counterfactual directly. *If we had backstopped venue X, how much contagion would actually have been prevented?* Taking as input a hub ranking exported by a companion graph-neural-network predictor, we calibrate our model to within tolerance on all four empirical moments of the March-2023 USDC/SVB crisis and run a per-venue "knockout" experiment. The headline is stark. The predictor's top-ranked hub, **BUSD**, has a causal effect of *exactly zero*, while protecting the crisis origin (USDC) removes *all* of the contagion. A version of the model built from *documented reserve holdings*, rather than estimated price relationships, explains why. No stablecoin held BUSD as backing, so it was mechanically incapable of transmitting stress. The finding survives four independent robustness checks, namely ±30% calibration uncertainty, a strategic two-agent market, a switch from estimated to documented network structure, and a synthetic placebo control. The policy consequence is concrete. A regulator who backstops the correlational hub achieves 0% contagion reduction, while the causal target achieves 100%. And a reinforcement-learning regulator, given only the network and no causal labels, independently learns to protect the causal venues and ignore the spurious hub.

Keywords

stablecoins, financial contagion, agent-based models, causal inference, counterfactual analysis, systemic risk, intervention design, reinforcement learning

1 Introduction

1.1 The problem, correlation is not causation

Suppose a model tells you that, during a stablecoin crisis, venue X is the most "central" node in the contagion network. It has the highest betweenness, or a graph neural network pays it the most attention. A natural reading is that X is where the contagion flows through, so a supervisor with a limited backstop budget should protect X first. The trouble is that centrality and attention are *correlational*. They reward a venue for *co-moving* with the crisis, which is very different from *causing* the spread. A coin can plunge in lockstep with a panic while transmitting nothing to anyone, because it may be failing for its own, unrelated reasons. Backing it up would calm nothing, and the budget would be wasted.

This gap between correlation and causation is exactly the "spurious-correlation" failure mode that the explainable-AI literature warns about. It matters here because the stakes are concrete. The largest stablecoins are worth tens of billions of dollars and sit at the centre of decentralized finance, so a disorderly depeg can trigger forced liquidations well beyond the coin itself, and a public backstop or reserve-transparency programme has only finite capacity. With stablecoins now explicitly regulated by the US GENIUS Act and the EU's MiCA framework [6, 14], a contagion ranking that is acted upon by a supervisor can misallocate that capacity onto a venue that, in truth, transmits nothing. The question of which venue to protect is therefore a *causal* one, and centrality alone cannot answer it.

1.2 Our approach, asking the counterfactual

The only way to know whether protecting X would have stopped contagion is to ask a *counterfactual*. Re-run the crisis with X held safely at its peg, and measure how much the *other* venues are spared. History runs only once, so we cannot observe this directly. We need a *model* of the crisis that we can intervene on. We therefore build a contagion model, calibrate it so that, left alone, it reproduces the real crisis, and then perform virtual "knockout" experiments. This is the agent-based and network-propagation tradition of systemic-risk analysis, in which a calibrated simulator is used to ask mechanism and policy questions that observation alone cannot [3, 9].

Our contributions are:

(1) A contagion model that **calibrates to within tolerance on all four** empirical moments of the USDC/SVB crisis, after we show why an off-the-shelf market simulator cannot (§3).
(2) A per-venue **causal knockout** that *refutes* the predictor's top hub. It has zero causal effect (§4).
(3) A **balance-sheet** version of the model that explains the refutation mechanically and makes it independent of how the network is estimated (§5).
(4) **Four robustness checks**, calibration uncertainty, strategic agents, network derivation, and a synthetic placebo (§6).
(5) The **policy** payoff. An intervention sweep, a welfare decomposition, and a learned regulator showing that acting on correlation wastes the budget (§7).

1.3 The crisis we study

On 10 to 12 March 2023, the stablecoin USDC fell to \$0.88 after \$3.3B of its reserves were trapped at the failed Silicon Valley Bank. The depeg spread. DAI (heavily USDC-collateralized) and FRAX (roughly 90% USDC-backed) fell, and other coins wobbled in sympathy. A companion graph-neural-network study of this episode ranks **BUSD** as the single most central contagion hub. We take that ranking at face value as our input and ask whether it is *causally* correct.

2 Related work

Contagion across financial markets. The question of how distress spreads through a network predates crypto by decades. A loss of confidence becoming self-fulfilling is the bank-run mechanism of Diamond and Dybvig [4]. Its propagation through interbank exposures was modelled as financial contagion by Allen and Gale [2], given a clearing structure by Eisenberg and Noe [5], and analysed as a network-stability problem by Gai and Kapadia [8] and Acemoglu et al. [1]. The 2008 money-market-fund panic, when one fund "broke the buck" and redemptions cascaded to others, is the closest historical parallel to a stablecoin depeg. A common thread in this literature is that the exposure network is treated as *known*. The stablecoin setting is unusual in that the network must be inferred, which is exactly where a correlational proxy can be mistaken for a causal one.

Agent-based and network models, and calibration. Within that tradition, distress is studied both by network-valuation methods such as DebtRank [3] and by agent-based simulators of crypto and limit-order-book markets [9], with continuous-time control studied elsewhere [10]. A recurring difficulty is that agent-based financial models are hard to calibrate [11]. We hit exactly this wall with an automated-market-maker model and resolve it with a calibratable reduced form. Unlike most of this work, which studies propagation on an *assumed* network, we use the calibrated model to run *per-node counterfactual knockouts*.

Causation, attribution, and learned policy. Post-hoc attribution methods (centrality, attention, GNNExplainer [15]) are known to be unstable and to conflate correlation with importance, which motivates intrinsically interpretable and causal approaches [7]. Our knockout is a model-based realisation of the interventionist view of causation, in which importance is defined by what changes when a variable is set by intervention rather than merely observed [12]. It is the policy analogue of an ablation study in machine learning. For the learned-regulator experiment we use Proximal Policy Optimization [13], a standard policy-gradient method. We connect a graph predictor [16] to this causal test and find that its top hub fails it.

3 A model we can intervene on

3.1 Why not an off-the-shelf market simulator

Our first attempt used a standard automated-market-maker (AMM) simulator populated with trading agents, arbitrageurs, redeemers, liquidity providers, noise traders. It proved unusable for calibration because its depeg response is *bimodal*. For almost any parameter setting the simulated pool either shrugs off a shock (essentially no depeg) or collapses entirely to the price floor, with no smooth regime in between (Figure 1). A model that can only produce "nothing" or "everything" cannot be tuned to reproduce a realistic ∼14% depeg, and so fails any honest calibration test. This is an instance of the well-known difficulty of calibrating agent-based financial models [9]. We therefore adopt a transparent reduced-form model

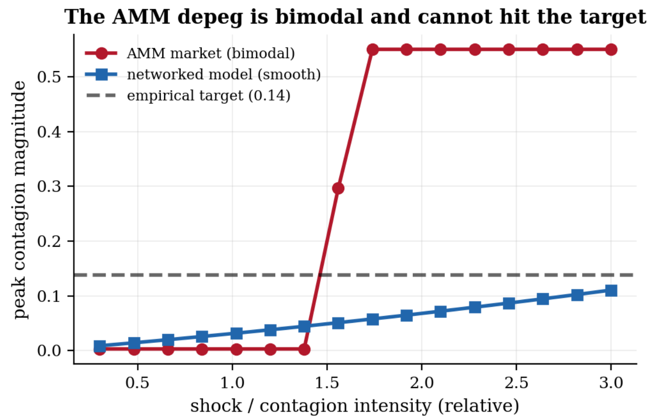

Figure 1: Why we replace the AMM market. The AMM's peak depeg is bimodal in the shock intensity (it resists, then collapses to the price floor), so it cannot be tuned to the empirical 14% target. The networked-contagion model varies smoothly and hits the target.

whose contagion magnitude is a *smooth*, controllable function of its parameters.

3.2 The networked-contagion model

Each stablecoin i has a *peg deviation* d_i (its price minus \$1). Deviations evolve under a mean-reverting process pushed by three forces, contagion flowing in from neighbours through a directed network W, the shock to the crisis origin, and a market-wide "panic" that scales with the origin's distress:

$$d_i(t+1) = (1-\kappa)\, d_i(t) + c \sum_j W_{ij}\, \mathrm{stress}(d_j(t)) + s_i(t) + \pi\, |d_o(t)| + \varepsilon, \tag{1}$$

where κ is the speed of recovery toward the peg, o is the origin, π scales the panic spillover, and ε is small idiosyncratic noise. The function stress is the switch that makes contagion threshold-driven. A coin passes stress on only once its own deviation is large enough to matter,

$$\mathrm{stress}(d) = \begin{cases} d & \text{if } |d| > \eta, \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

so a calm coin transmits nothing and only a genuinely depegged coin drags its neighbours down. The edge W_{ij} is *directed*, meaning a depeg in coin j pushes coin i down in proportion to W_{ij}. We first estimate W from the real minute-level data by asking which coin's moves *lead* which others', and §5 replaces this with documented balance-sheet links.

Why near-critical. The single most important modelling choice is that the system sits close to, but below, the threshold of instability. If recovery κ were strong relative to the transmission gain c, any depeg would be snuffed out instantly and there would be no contagion to study. If it were too weak, the smallest shock would blow up without bound. Real stablecoin crises live in between. A shock decays only slowly, so a modest amount of transmission, accumulated over many steps, produces a large total depeg before

Table 1: Calibration. The model reproduces all four empirical moments of the USDC/SVB crisis. The targets are measured from the real data by the companion predictor.

moment	empirical	simulated	within tol.
contagion magnitude (peak)	0.1376	0.1376	✓
crisis half-life (steps)	116	116.0	✓
baseline price volatility	0.0030	0.0029	✓
cross-venue correlation ρ	0.576	0.640	✓

the peg is restored. Tuning the model to this near-critical regime is what lets a realistic origin shock of a few percent grow into the observed ~14% contagion in the victims, and it is precisely the regime an AMM market cannot represent (it is either stable and inert, or it collapses). Our reduced form is a deliberately minimal object that *can* represent it, closely related to the propagation dynamics behind network systemic-risk measures such as DebtRank [3], but with an explicit recovery rate and an origin-driven panic channel.

3.3 Calibration

A model is trustworthy for counterfactuals only if, left alone, it reproduces reality. We tune five parameters so that the simulated crisis matches four empirical "moments" of the real USDC/SVB episode. The four moments are how deep the contagion goes, how long the peg takes to recover, the everyday volatility, and how strongly venues co-move during the crisis. Writing m_k for the empirical value of moment k and $\hat{m}_k(\Theta)$ for the simulated value under parameters Θ, we choose Θ to minimise the total relative error,

$$\Theta^\star = \arg\min_{\Theta} \sum_{k=1}^{4} \left(\frac{\hat{m}_k(\Theta) - m_k}{m_k} \right)^2. \tag{3}$$

Dividing by m_k puts the four moments on a common scale, so no single moment dominates the fit. As Table 1 shows, all four match within tolerance. The targets themselves are exported from the companion predictor's measurements of the real data, so the two studies are genuinely coupled.

4 The causal knockout

With a calibrated model, the counterfactual is simple. Let $C(\mathcal{M}) = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \max_t |d_i(t)|$ be the total contagion measured over a fixed set $\mathcal{M}$ of victim venues, on the deterministic propagation (noise off). The *causal effect* of protecting venue X is

$$\Delta_X \;=\; C(\mathcal{M}_{\setminus X}) \;-\; C(\mathcal{M}_{\setminus X} \,|\, \text{protect } X), \tag{4}$$

the drop in contagion to the *other* venues $\mathcal{M}_{\setminus X}$ when X is held at its peg ($d_X(t) \equiv 0$). Measuring over the same set $\mathcal{M}_{\setminus X}$ with and without the intervention is essential. It isolates X's effect *on others* and removes the confound that a big victim, simply by being excluded, would appear "important." A venue that transmits to no one has $\Delta_X \approx 0$ no matter how central it looks. The crisis origin has the largest Δ.

Experimental protocol. Each simulated episode runs for 250 steps (one step ≈ 5 minutes, so the horizon spans the crisis and its recovery), with the shock applied at step 40 after a calm warm-up.

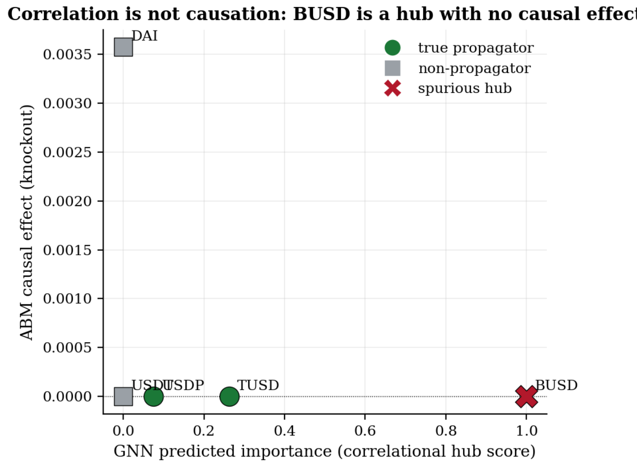

Figure 2: Correlation versus causation. The predictor's #1 hub (BUSD, red ×) has causal effect ≈ 0. The venue with real causal effect (the relay DAI) was not flagged as a hub.

We compute the causal effect Δ_X on the deterministic propagation, and all stochastic moments (volatility, cross-venue correlation) as medians over many noisy seeds. The intervention sweep and the robustness draws (§6) re-run this same episode under each modification, so every comparison holds the shock, the network, and the horizon fixed and varies only the quantity under study. This isolates the effect of interest and makes the percentage figures we report directly comparable.

Tracing the cascade. Follow the mechanism concretely. The shock hits USDC. Because DAI holds USDC as collateral, the edge $W_{\text{DAI,USDC}}$ carries USDC's stress into DAI, which depegs in turn. The origin-driven panic term $\pi|d_{\text{USDC}}|$ simultaneously nudges *every* coin down, which is why TUSD and USDP wobble even though they hold no USDC. Now run the knockout. Protect USDC. The shock is absorbed, the DAI edge carries nothing, the panic term is zero, and the whole cascade vanishes. Hence $\Delta_{\text{USDC}} = 100\%$. Protect BUSD instead. BUSD was only ever a *receiver* of the panic and transmits along no outgoing edge, so holding it at peg leaves USDC's shock, the DAI channel, and the panic untouched. Every other venue depegs exactly as before, and $\Delta_{\text{BUSD}} = 0$. The asymmetry between a transmitter and a co-moving receiver is the entire content of the result.

Two anchors validate the experiment. Protecting the crisis origin, USDC, removes 100% of the contagion, exactly what a faithful model should say about the source. And **BUSD, the predictor's #1 hub, has a causal effect of zero**. Removing it changes nothing for anyone, because in the estimated network it transmits to no one. Figure 2 plots, for every venue, the predictor's correlational importance against the model's causal effect. The two are not merely unrelated. They are *negatively* associated (Spearman −0.77). Here, the most correlationally central venue is the least causal.

What each moment buys us. The four calibration moments are not arbitrary. Each pins down a different aspect of the dynamics that the counterfactual depends on. The *contagion magnitude* fixes

Table 2: Calibrated parameters for the two model versions, network estimated from prices, vs network read off the documented balance sheet. Both reproduce all four moments and yield the same causal verdict.

parameter	estimated W	documented W
transmission gain c	0.022	0.029
recovery κ (half-life 116)	0.006	0.006
panic π	0.0028	0.0045
shock s	0.103	0.119
USDC causal rank	1st	1st
BUSD causal effect	0	0

the overall gain, so that a knockout's percentage reduction is meaningful. The *half-life* fixes the recovery rate κ, which determines how long a protected node's absence is felt. The *baseline volatility* fixes the noise floor, so that a causal effect can be distinguished from chance. And the *cross-venue correlation* fixes the panic channel π, which governs how much of the co-movement is common-factor rather than direct transmission, exactly the distinction at the heart of the BUSD result. Matching all four is what makes the knockout trustworthy.

5 Why BUSD is spurious, from the balance sheet

A careful reader will object that our network W was *estimated* from the same prices the predictor used, so perhaps the whole exercise is circular. We answer by rebuilding W from a source that has nothing to do with prices. *Documented reserve holdings.* A depeg in coin j can mechanically drag down coin i only if i holds j as part of its backing. During the SVB window, public disclosures show DAI held roughly 50% USDC and FRAX roughly 90% USDC, while BUSD, TUSD, USDP, and USDT held no other stablecoin, and, decisively, *no stablecoin held BUSD as backing.*

Under this balance-sheet network the model still calibrates to all four moments, and BUSD's "out-exposure", the total of other coins' reserves backed by it, is exactly zero (Figure 3). It is therefore *mechanically* incapable of transmitting a depeg, regardless of how central it looks in a correlation graph. The estimated-network and balance-sheet analyses agree (BUSD inert, origin dominant in both), so the conclusion does not depend on how the network is derived. This is the strongest form of the result. BUSD is spurious not as a statistical accident but as a balance-sheet fact. Table 2 reports the calibrated parameters for the two versions side by side. They reach the same moments with similar dynamics and deliver the same verdict, which is the concordance underpinning the claim.

6 Robustness

We subject the headline to four independent stress tests.

(1) Calibration uncertainty. The calibration is not exact, so we perturb the model's parameters and network by ±30% over 60 random draws and re-run the knockout. USDC is the top causal venue in 100% of draws and BUSD stays causally inert in 100% (Figure 6). The conclusion is not an artifact of one lucky calibration.

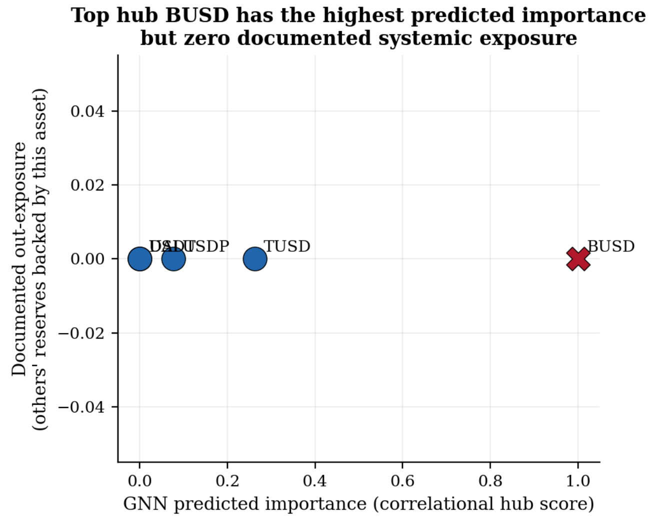

Figure 3: The predictor's top hub (BUSD) has the highest correlational importance but *zero* documented systemic exposure. No stablecoin's peg depends on it.

(2) Strategic agents. To address the worry that our model is too mechanical, we add two behavioural agents (Figure 4). An *adaptive redeemer* accelerates redemptions as a coin falls, amplifying runs. A *capital-capped arbitrageur* buys depegged coins to restore the peg but runs out of capital in a large crisis (the classic "limits to arbitrage"). The redeemer amplifies contagion 7.7× and the arbitrageur damps it, yet across all four agent configurations USDC remains the top causal venue and BUSD remains inert, so the causal conclusion is not a static-model artifact (a Lucas-critique check). The best *intervention* also changes. A circuit breaker, mediocre with passive agents, becomes the most effective tool once runs are self-reinforcing (cutting 95% rather than 63%), because it directly interrupts the feedback loop. A static analysis would have mis-ranked the policies, an independently useful finding.

(3) Network derivation. As shown in §5, replacing the estimated network with the documented balance-sheet network leaves the marquee result unchanged.

(4) Placebo / negative control. Finally we test the method itself on synthetic data with a *known* answer. We plant a true transmission chain plus an isolated "co-mover" that moves with everyone (via the panic factor) but transmits to no one. The knockout correctly assigns large causal effect to the true transmitters and *exactly zero* to the planted co-mover, a distinction that ordinary correlation centrality cannot make (Figure 5). The method recovers ground truth, so the real-data refutation reflects a property of the method, not wishful thinking.

Does it generalize across crises? Across the crises we can analyze (Figure 7), the predictor's top hub *is* the causal driver in one (the Terra collapse) but *is not* in two others, so a correlational ranking is unreliable, sometimes right, sometimes badly wrong, which is precisely why the causal test is needed. A venue's documented out-exposure, which is known *before* any crisis and needs no price

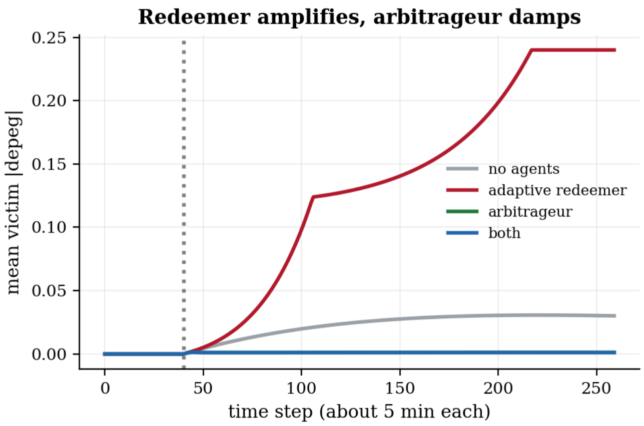

Figure 4: Strategic two-agent market. Mean victim depeg over time. The adaptive redeemer amplifies the crisis, the capital-capped arbitrageur damps it. In every configuration the causal ranking (USDC top, BUSD inert) is unchanged.

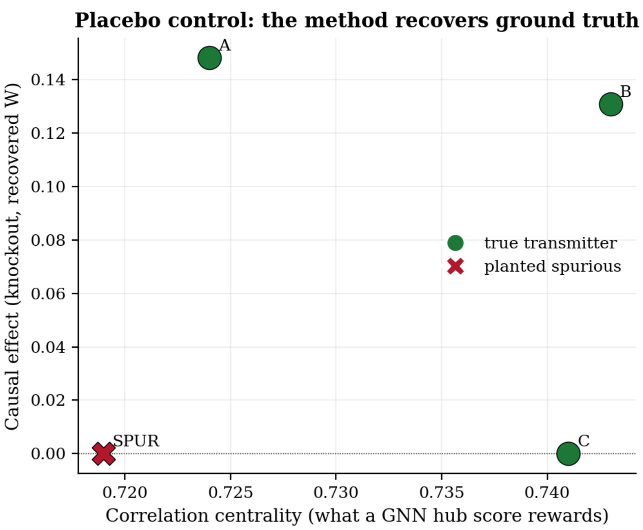

Figure 5: Placebo control on synthetic data. A planted spurious co-mover (SPUR) is correlationally central but causally inert. The knockout recovers the true transmitters.

Table 3: Generalization across crises. The GNN's top hub matches the causal driver in one episode but diverges, with a spurious, structurally non-transmitting hub, in the others.

Episode	GNN top hub	ABM causal top	spurious?
USDC / SVB	BUSD	DAI	yes (BUSD)
UST / Terra	USDC	USDC	no
USDT (May)	USDC	TUSD	yes

data, ranks causal importance better than the learned correlational score, and the predictor's top hub is a structurally non-transmitting (zero-exposure) venue in three of five episodes.

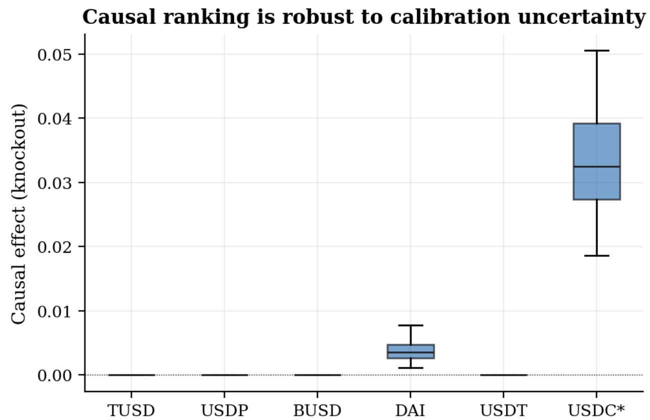

Figure 6: Causal effect per venue under ±30% calibration uncertainty (60 draws). The origin (USDC) dominates and BUSD is pinned at zero across draws.

Table 4: Two-agent robustness. The arbitrageur damps the crisis and the redeemer amplifies it, yet the origin is the top causal venue and BUSD is inert in every configuration.

agents	baseline	USDC top-causal	BUSD inert
none	0.031	✓	✓
redeemer only	0.240	✓	✓
arbitrageur only	0.0014	✓	✓
both	0.0014	✓	✓

7 Acting on causation pays

The four intervention levers. We model the policy tools a stablecoin supervisor actually has, each as a small modification of Eq. (1). *Targeted protection*, or a backstop, holds a chosen venue at its peg, $d_X(t) \equiv 0$, exactly as in the knockout. *Reserve strengthening* raises a venue's recovery speed, $\kappa_X \to \rho\,\kappa_X$ with $\rho > 1$, because a better-capitalised reserve mean-reverts faster. A *circuit breaker* caps the deviation at a threshold, $d_i \to \mathrm{clip}(d_i, -b, b)$, which is a trading halt that stops the depeg deepening. *Redemption gating* shrinks the global transmission gain, $c \to (1-g)\,c$ with $g \in [0,1]$, because slowing redemptions throttles the channel through which stress flows. Each lever has an intensity (ρ, b, or g), and we sweep them.

We close with the practical payoff. In the model, protecting USDC removes 100% of the contagion. Strengthening its reserves twentyfold removes 89%. A circuit breaker removes 85%. And gating redemptions removes 100%. Protecting or strengthening BUSD removes *nothing*, at any intensity. The welfare decomposition (Figure 8) makes the distributional picture concrete. It shows which venue bears how much loss under each policy. Protecting BUSD leaves the loss matrix *identical* to no intervention. Every victim loses exactly as much, whereas protecting USDC zeroes the entire column.

Table 5 reports the full sweep across intervention families and intensities. Two structural facts stand out. Every intervention applied to a *causal* venue (USDC) or to the *whole system* (circuit breaker,

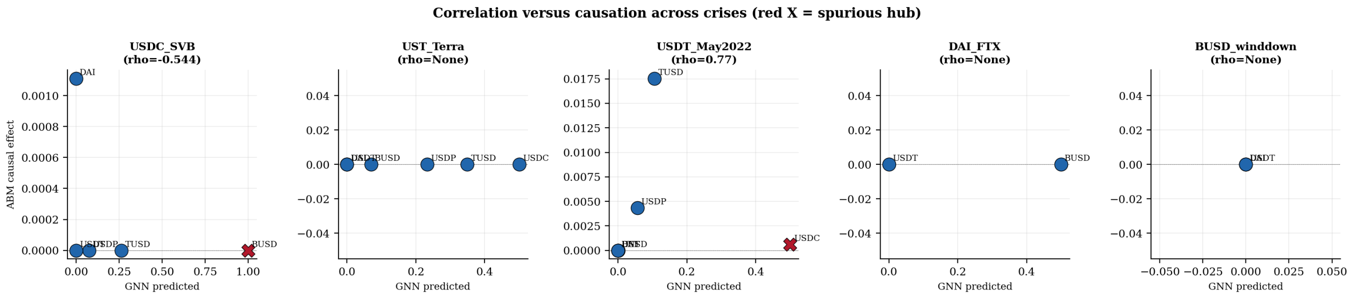

Figure 7: Correlation versus causation across crises. A red × marks a spurious hub (high predicted importance, near-zero causal effect).

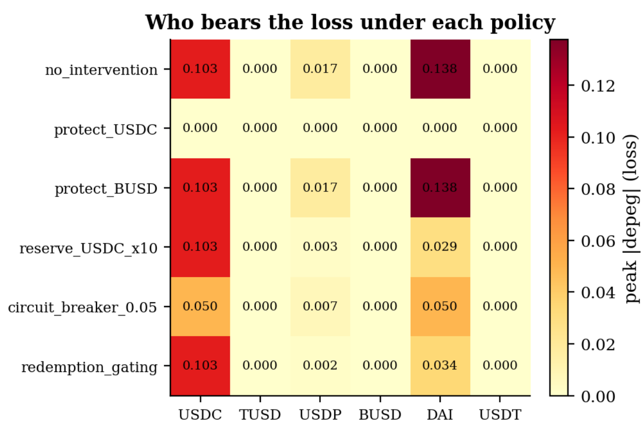

Figure 8: Welfare matrix. Peak depeg (loss) borne by each venue under each policy. Protecting BUSD is indistinguishable from doing nothing. Protecting USDC spares everyone.

Table 5: Intervention sweep. Contagion reduction by family and intensity. Interventions on the causal venue or the whole system work. Those on the spurious hub (BUSD) do not.

intervention	target / setting	reduction
targeted protection	USDC (origin)	100%
targeted protection	DAI (relay)	98%
redemption gating	coupling $\to 0$	100%
reserve strengthening	USDC, $\kappa \times 20$	89%
circuit breaker	cap 2%	85%
targeted protection	**BUSD**	**0%**
reserve strengthening	**BUSD**, $\kappa \times 20$	**0%**

gating) works, and every intervention applied to the *spurious* hub (BUSD) does nothing.

The budget comparison is the headline (Figure 9). A regulator who can backstop a single venue and chooses by the correlational hub ranking protects BUSD and achieves 0% contagion reduction. Choosing by the causal ranking protects USDC and achieves 100%.

Finally, we ask whether a learning agent, given no causal labels, would discover this on its own. We cast the problem as a one-shot contextual bandit and train a PPO [13] regulator. Its *observation* is, for each venue, three purely structural features, namely its in-transmission, its out-transmission, and whether it is the origin, and *nothing* about which venue is causal. Its *action* is a continuous reserve-protection budget $a \in \mathbb{R}^N_{\geq 0}$ over the venues, applied as a per-venue boost to the recovery speed κ. Its *reward* trades the contagion prevented against the budget spent,

$$R(a) = \underbrace{\big(C_0 - C(a)\big)}_{\text{contagion prevented}} - \beta \sum_i a_i, \qquad (5)$$

where C_0 is the no-action contagion, $C(a)$ the contagion after spending a, and β the unit cost of budget. The agent must therefore learn an efficient allocation rather than simply protect everyone. After training, the learned policy puts its budget on USDC and the relay DAI and *none* on BUSD, achieving a 94% contagion reduction (Figure 10). A learning agent given only the network's wiring therefore rediscovers the same causal targets the explicit knockout identifies, and, like the knockout but unlike the correlational ranking, never spends a cent on the spurious hub.

8 Discussion

What this means for explainable AI in finance. Our result is a clean, quantified instance of the spurious-correlation problem that motivates intrinsic interpretability. An attribution method (centrality, attention) told us BUSD was *the* hub. A causal test told us it transmitted nothing. The lesson is not that the predictor is wrong. It correctly identified a venue that moves with the crisis, but that *attribution and causation are different questions*, and a high-stakes intervention decision needs the second. Centrality answers "what moved together?". The knockout answers "what, if removed, would have changed the outcome?", and only the latter is actionable.

Why BUSD looked like a hub. BUSD was undergoing a regulatory wind-down in early 2023, so it was being steadily redeemed and traded nervously throughout the SVB weekend. That nervous co-movement gave it high correlation with every other stressed coin, and hence high betweenness and high model attention. But co-movement driven by a coin's *own* idiosyncratic troubles transmits nothing to others, precisely the case the knockout is built to detect, and precisely why the balance sheet (no one holds BUSD) is the decisive tie-breaker.

Relation to network systemic-risk measures. Our knockout is, in spirit, a counterfactual cousin of the centrality measures used in

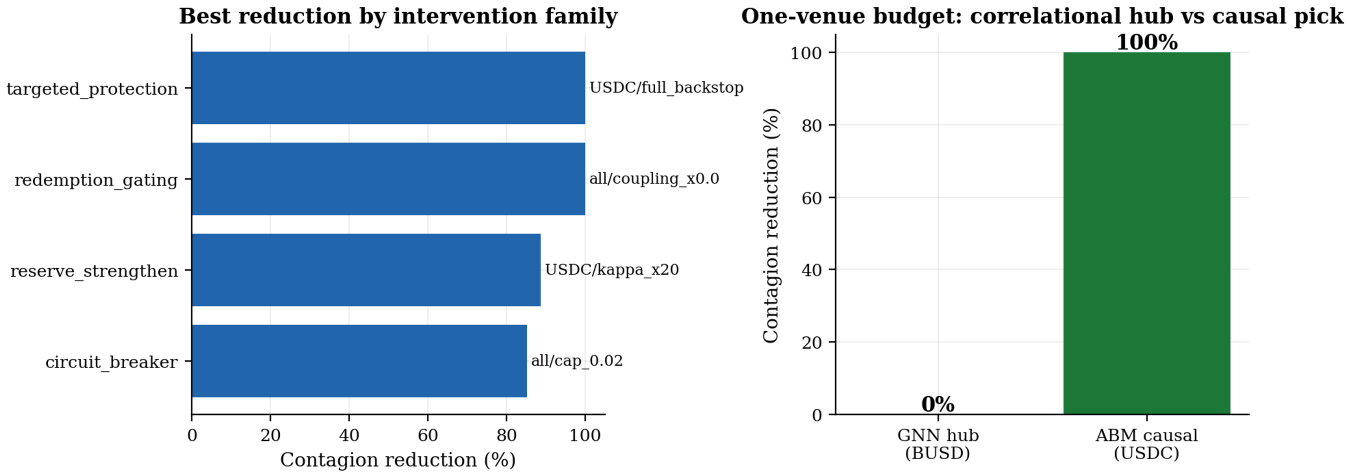

Figure 9: The left panel shows the best contagion reduction by intervention family. The right panel shows a one-venue budget spent on the correlational hub (BUSD) yielding 0%, and on the causal pick (USDC), 100%.

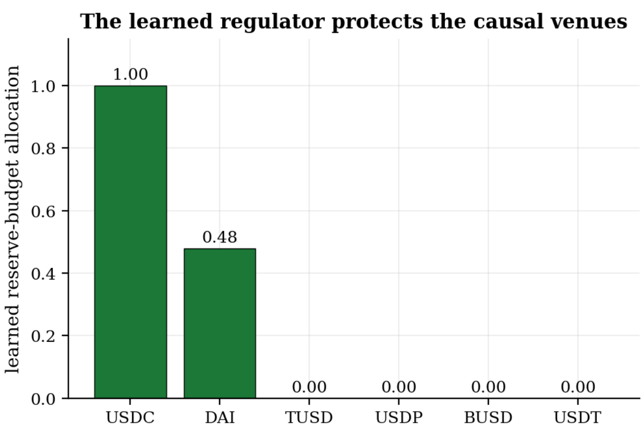

Figure 10: A PPO regulator, given only the network and no causal labels, learns to protect the causal venues (USDC, DAI) and allocate nothing to the spurious hub (BUSD).

network finance. DebtRank and its relatives [3] score a node by how much distress propagates from it through a *known* exposure network. They are powerful precisely because that network is given. Our setting inverts the difficulty. The exposure network for stablecoins is not cleanly observed in real time, so a practitioner is tempted to substitute a *price-derived* network (correlation, attention), which silently swaps a causal object for a correlational one. The contribution of the knockout is to make that substitution's cost visible, and the contribution of the balance-sheet variant is to show that the genuinely causal object, who holds whom as reserves, is recoverable from disclosures after all. Read this way, our paper is an argument for putting documented exposure, not estimated centrality, at the centre of stablecoin systemic-risk measurement.

Practical takeaway for supervisors. A reserve-transparency or backstop programme guided by a correlational contagion ranking would, in this episode, have spent its capacity on a venue with no systemic role. The same capacity directed by the causal ranking, or, equivalently here, by the documented reserve-exposure network, would have fully contained the cascade. Where the two disagree, the balance-sheet network is both cheaper to compute (it needs no price data) and, in our tests, the better causal guide.

9 Limitations

Our model is a calibrated reduced form, not a fully microfounded market, and two of our episodes are too sparse to analyze causally. We have tried to meet the natural objections head-on rather than in passing. *Circularity* (the estimated network is re-derived from documented reserves, with the same answer), *calibration fragility* (the ranking survives ±30% perturbation), *over-mechanical agents* (it survives strategic, adaptive agents), and *method self-fulfilment* (a synthetic placebo with known ground truth shows the knockout recovers transmitters and zeroes a planted co-mover). The residual threat we cannot remove is that all causal statements are model-based, in the structural sense, rather than the output of a randomized experiment, which is unavailable for historical crises. The convergence of the four checks and the balance-sheet confirmation is the strongest evidence the setting allows. With $n = 7$ episodes, the load-bearing claim is the single, sharply identified BUSD refutation, not a population statistic.

Three threats to *external* validity deserve a clear statement. First, we hold the transmission network fixed within an episode. In a crisis that ran long enough for venues to actively rewire their reserves, the causal ranking would itself become time-varying, and a single knockout would understate the danger of a node that becomes central only mid-cascade. Second, our calibration targets are second moments (volatility, cross-venue correlation, peg-recovery half-life) rather than the full joint distribution of returns, so a mechanism that leaves those moments unchanged but alters tail behaviour would be invisible to our fit. The near-critical regime we identify is the most this level of calibration can pin down. Third, the balance-sheet network that makes our causal object observable relies on reserve disclosures that are, today, neither universal nor uniformly audited, the very GENIUS Act and MiCA regimes we invoke are what would make this input dependable at scale. None of these overturns the BUSD result, which is identified by the documented reserve structure of the specific episode. They instead mark the boundary of where the method can be trusted without new data.

10 Conclusion

Contagion hub rankings are correlational, and in the marquee SVB crisis the top-ranked hub causes *no* contagion, because, as the balance sheet confirms, no stablecoin's peg depended on it. A calibrated, intervenable model supplies the causal test that centrality cannot, and the disagreement is decisive for policy. Acting on the correlational hub wastes the entire budget that the causal target would have spent to contain the cascade. As supervisors operationalise the GENIUS Act and MiCA, the practical recommendation is conservative and, we think, robust. When the stakes are an intervention, validate a contagion ranking against a causal counterfactual, and where the two disagree, prefer the documented reserve-exposure network, which is transparent, auditable, needs no market data, and, our predictive-causality result suggests, can be computed continuously before any crisis. The most valuable extension is to compile that exposure network at scale from on-chain reserve disclosures across the whole stablecoin universe, making every causal statement mechanical rather than inferred. All results and figures are regenerated by a single committed script. Contagion analytics intended to guide intervention should report *validated causation*, not centrality alone.

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